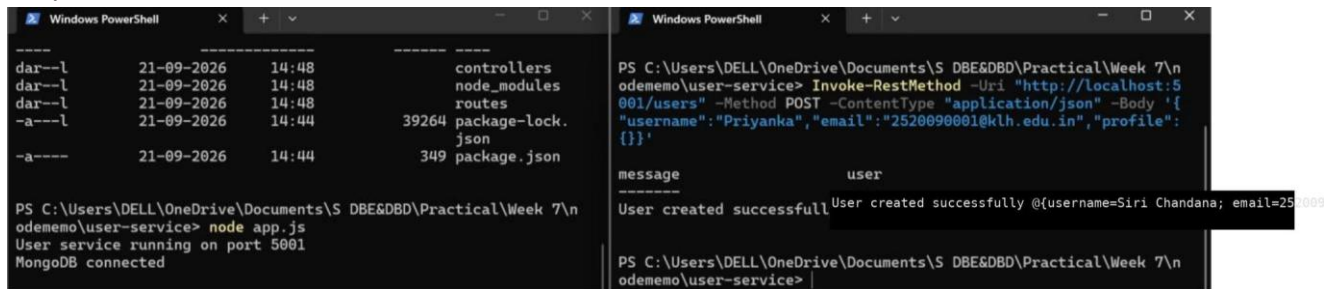


WEEK-7 PRACTICAL — Node.js User Service

[Siri Chandana, 2520090198]

Step — 1 Create a User — POST



```
-----
dar--l      21-09-2026    14:48      controllers
dar--l      21-09-2026    14:48      node_modules
dar--l      21-09-2026    14:48      routes
-a---l      21-09-2026    14:44      39264 package-lock.
-a-----   21-09-2026    14:44      349 package.json

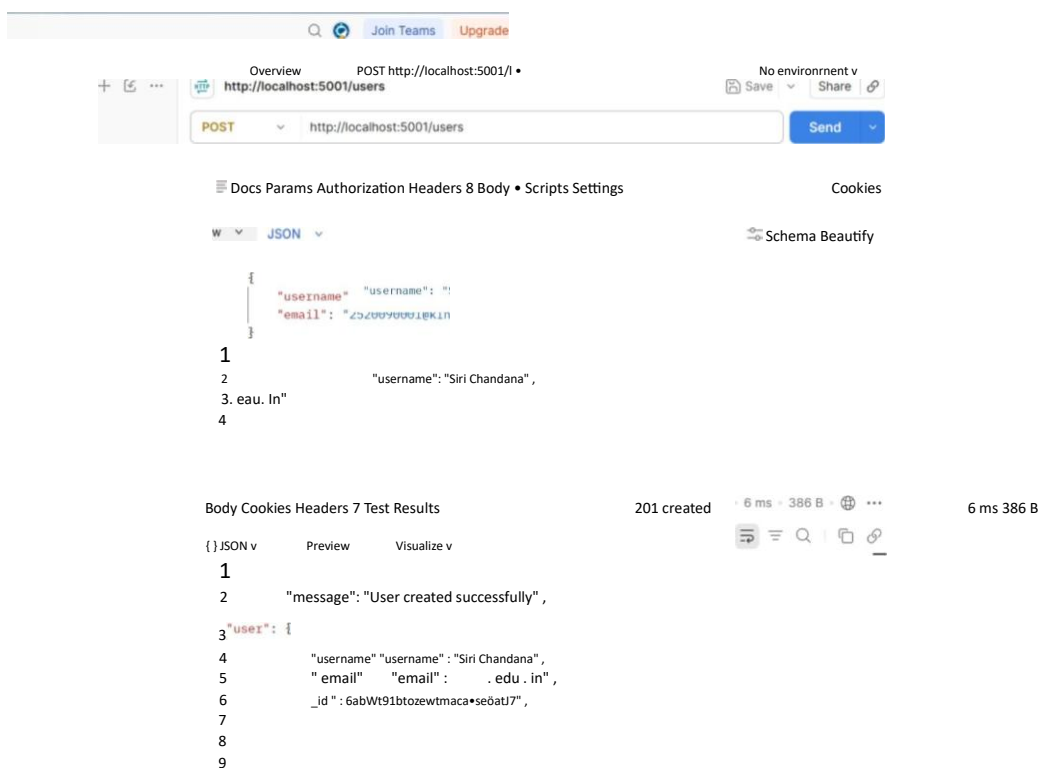
PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7\n
odememo\user-service> node app.js
User service running on port 5001
MongoDB connected

PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7\n
odememo\user-service> Invoke-RestMethod -Uri "http://localhost:5
001/users" -Method POST -ContentType "application/json" -Body '{
"username":"Priyanka","email":"2520090001@klh.edu.in","profile":
{}}'

message          user
-----
User created successfull  User created successfully @{username=Siri Chandana; email=2520090198@klh.edu.in; profile={}}

PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7\n
odememo\user-service>
```

POST in mongosh:



Overview POST http://localhost:5001/users

POST http://localhost:5001/users

JSON

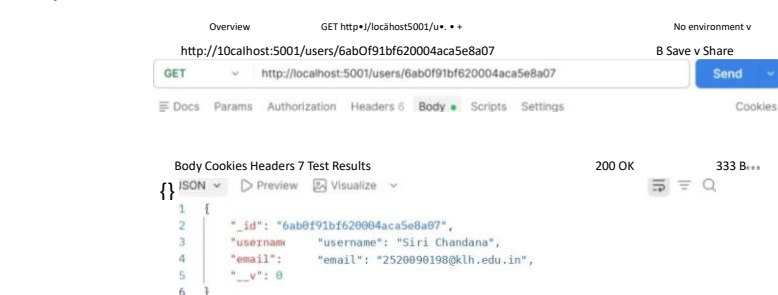
```
{
  "username": "Siri Chandana",
  "email": "2520090198@klh.edu.in"
}
```

201 created 6 ms 386 B

Body Cookies Headers 7 Test Results

```
{
  "message": "User created successfully",
  "user": {
    "username": "Siri Chandana",
    "email": "2520090198@klh.edu.in",
    "_id": "6ab0f91bf620004aca5e8a07"
  }
}
```

Step 2: Get User — GET



Overview GET http://localhost:5001/users/6ab0f91bf620004aca5e8a07

GET http://localhost:5001/users/6ab0f91bf620004aca5e8a07

JSON

```
{
  "_id": "6ab0f91bf620004aca5e8a07",
  "username": "Siri Chandana",
  "email": "2520090198@klh.edu.in",
  "_v": 0
}
```

200 OK 333 B

Output on browser by id:

O localhost: 5001 /users/6ab0f91 bf620004aca5e8a07

Pretty print a

```
{
  "_id": "6ab0f91bf620004aca5e8a07",
  "username": "Siri Chandana",
  "email": "email"  "2520090198@k1h.edu.in",
  "__v": 0
}
```

Step 3: Create Activity — POST

```
PS C:\Users\DELL\OneDrive\Documents\S DBE5DBD\Practical\Week 7\nodememo\user-service> Invoke-RestMethod -Uri "http://localhost:5001/a
ctivities" -Method POST -ContentType "application/json" -Body '{"user_id":"6ab0f91bf620004aca5e8a07","action":"SEARCH_PYTHON"}'

message          activity
-----
Activity created successfully @{user_id=6ab0f91bf620004aca5e...
```

In mongosh:

KAMTAM PRIYANKA's Workspace

Join Teams Upgrade

Overview

POST http://localhost:5001/s

No environment

+

🔗

...

🔗

http:localhost:5001/activities

POST

v http:localhost:5001/activities

🔗

Save v

Share

Send

Docs Params Authorization Headers B Body • Scripts Settings

Cookies

Schema Beautify

JSON v

1

2"user_id "6abOf91bf620004aca5e8a07

3,"

4"action ": "SEARCH_JAVA"

5

Body Cookies Headers 7 Test Results

201 created

439 B

{ } JSON v

Preview

Visualize v

-p

1

2"message": "Activity created successfully",

3"activity": {

4"action": "SEARCH_JAVA",

5"user_id": "6ab0f91bf620004aca5e8a07",

6"timestamp": "2026-09-21T09:40:46.119Z",

7"__v": 0

8}

9

10

Step 4: Add More Activities

KAMTAM PRIYANKAS workspace v

Q Search

Overview

POST http://localhost:5001/z • •

+

🔗

...

🔗

http:localhost:5001/activities/6abOf91bf620004acase8a07

POST

http:localhost:5001/activities/6abOf91bf620004aca5e8a07

🔗

Docs Params Authorization Headers 8 Body • Scripts Settings

O none C) form-data O x-www-form-urlencoded O raw

binary

1

```

{
  "user_id": "6ab0f91bf620004aca5e8a07",
  "activities": [
    {
      "action": "Perform Week 7 Skill",
    },
    {
      "action": "Perform Week 7 Practic",
    },
    {
      "action": "Complete MongoDB Task",
    },
    {
      "action": "Test APIs in Postman"
    }
  ]
}

```

2

Body Cookies Headers 7 Test Results

JSON Preview Visualize

```

1 [
2   {
3     "_id": "6ab10015f620004aca5e8a0c",
4     "user_id": "6ab0f91bf620004aca5e8a07",
5     "action": "SEARCH_PYTHON",
6     "timestamp": "2026-09-21T09:59:49.270Z",
7     "__v": 0
8   },
9   {
10    "_id": "6ab0fe44f620004aca5e8a09",
11    "user_id": "6ab0f91bf620004aca5e8a07",
12    "action": "Perform Week 7 Practical",
13    "timestamp": "2026-09-21T09:52:04.478Z",
14    "__v": 0
15  }
16 ]

```

```

PS C:\Users\DELL\OneDrive\Documents\5 DBE60BD\Practical\Week 7\nodememo\user-service> Invoke-RestMethod -Uri "http://localhost:5001/activities/6ab0f91bf620004aca5e8a07" -Method GET

_id      : 6ab10015f620004aca5e8a0c
user_id   : 6ab0f91bf620004aca5e8a07
action    : SEARCH_PYTHON
timestamp : 2026-09-21T09:59:49.270Z
__v       : 0

_id      : 6ab0fe44f620004aca5e8a09
user_id   : 6ab0f91bf620004aca5e8a07
action    : Perform Week 7 Practical
timestamp : 2026-09-21T09:52:04.478Z
__v       : 0

_id      : 6ab0fb9ef620004aca5e8a08
user_id   : 6ab0f91bf620004aca5e8a07
action    : SEARCH_JAVA
timestamp : 2026-09-21T09:40:46.119Z
__v       : 0

```

Output on browser:

a O local host: 5001 /activities/6ab0f91 bf620004aca5e8a07

Pretty print ☒ print

```

{
  "id": "6ab1e015f620004aca5e8a0c",
  "user id": "6ab0f91bf620004aca5e8a07",
  "action": "SEARCH_PYTHON",
  "timestamp": "2026-09-21T09:59:49.270Z",
  "__v": 0
},
{
  "timestamp": "2026-09-21T09:52:04.478Z",
  "id": "6ab0fe44f620004aca5e8a09",
  "user id": "6ab0f91bf620004aca5e8a07",
  "action": "Perform Week 7 Practical",
  "timestamp": "2026-09-21T09:52:04.478Z",
  "__v": 0
},
{
  "timestamp": "2026-09-21T09:40:46.119Z",
  "id": "6ab0fb9ef620004aca5e8a08",
  "user id": "6ab0f91bf620004aca5e8a07",
  "action": "SEARCH_JAVA",
  "timestamp": "2026-09-21T09:40:46.119Z",
  "__v": 0
}
]

```

MongoDB connection:

• MongoDBCompass- 12700.127017/digital_library

Compass

O My Queries

Data Modeling

Search connections

127.0.0.1:27017 > digital_library

Open MongoDB shell

Create collection

Refresh

Collection name	Properties	Storage size	Data size	Documents	Avg. document size
activities	-	32.77 kB	300.00 B	3	100.00 B
users	-	36.86 kB	339.00 B	4	84.00 B

CONNECTIONS

- 127.0.0.1:27017
 - BookFlow
 - College
 - admin
 - config
 - digital_library
 - activities
 - users

• MongoDB Compass -127.0.0.1:27017/digital_library.activities

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

- 127.0.0.1:27017
- BookFlow

College

- activities
- users
- local

e student_records

Dbsml

Week-3_proctical

- MongoDB Compass - 127.0.0.1:27017/digital_library.users

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

Welcome activities

127.0.0.1:27017 > digital_library > activities

Open MongoDB Shell

Documents 3 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or Generate query

25 1-3 of 3

_id: ObjectId('6abgfb9ef62eee4aca5e8ae5')

user_id : ObjectId('6abef91bf62ee0aacaseae7')

action : "SEARCH_JAVA"

admin

timestamp : ISODate('2026-09-21T09:48:46.119+00:00') 21 September 2026 at 09:48:46 UTC

config = digital_library

ObjectId('6ab0fe44f62ee0aacasesa09') user_id : ObjectId('6abef91bf620ee4aca5e8ae7') action:

"Perform Week 7 Practical " ti.esta.p : ISODate('2026-09-21T09:52:04.478+00:00') 21 September 2026 at 09:52:04 UTC

_id: ObjectId('6ab18615f620004aca5e8ae')

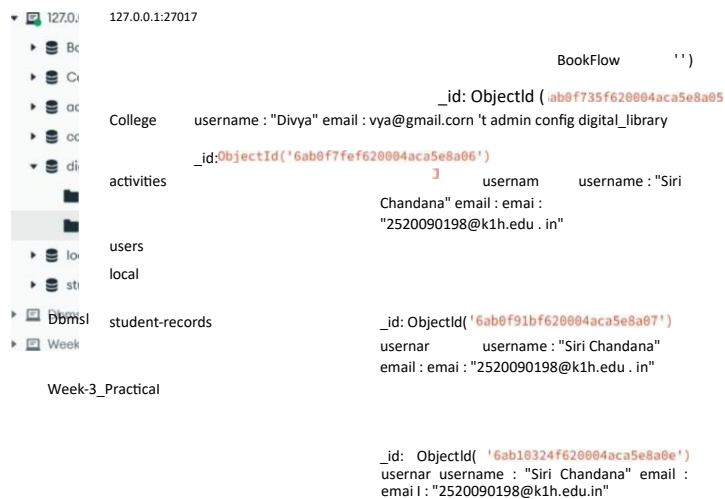
Week-3_proctical user_id : ObjectId('6abef91bfnoe@gaca508a07') action : "SEARCH_PVTHON" ti.esta.p : ISODate('2026-09-21T09:59:49.270+00:00') 21 September 2026 at 09:59:49 UTC

Welcome users

127.0.0.1:27017 > digital_library > users

Documents 4 Aggregations Schema Indexes 1 Validation

Type Explain



Folder Structure:

Week 7

I nodememo

| _ user-service

controllers

| 1— activityController.js

| 1— userController.js

routes

activityRoutes.js

| 1— userRoutes.js

1— app.js

db.js

1— package.json

└ .env

```
db.js const mongoose = require( "mongoose" );
```

```
const MONGO URI =
```

```
process.env.MONGO URI ||
```

```
"mongodb : / / 127.0.0.1 : 27017
```

```
/digital_library" ;
```

```
mongoose. connect(MONGO URI)
```

```

        . then(() console. log("MongoDB  connected")) . catch(err => console.
        error("MongoDB connection error: " , err) );

const userSchema = new mongoose. Schema({
    username: { type: String, required: true

    email:  {   type:
        String,
        required: true

    profile: { type: mongoose. Schema . Types .
        Mixed, default: { }

const activitySchema = new mongoose.
    Schema({ user_id: { type: mongoose.
    Schema . Types.ObjectId, required: true,
    ref: "User"

    action: {   type:
        String,
        required: true

    timestamp: { type: Date,
        default: Date. now

const User = mongoose. model ("User", userSchema); const
Activity = mongoose . model ( "Activity" , activitySchema) ;

module . exports{ User, Activity };

controllers/userController.js  const  {
User } = require(" . . / db");

const createUser -- async (req, res) => { try { const {
    username, email, profile } = req . body;

        const user = await User. create({
            username, email, profile

            res. status( 201).j message: "User
            created successfully", user

        } catch (error) { res.
            status(500) .json({

```

```
message : "Error creating user",
error: error. message
```

```
const getUser - async (req, res) => { try { const user = await
  User.findById(req.params.id);

  if ( ! user) return res. status(404). json(
    message . • "User not found"

    res.json(user);
  } catch (error) { res.
    status(500) .json({
      message : "Error getting user",
      error: error. message
```

```
module . exports   { createUser, getUser } ;
controllers/activityContr011er. js
```

```
const mongoose = require( "mongoose" ) ; const {
Activity } require( "../db"
```

```
const createActivity = async (req, res) => {try {
  const { user_id, action } - req . body;
  if
    (
      !mongoose.Types.ObjectId.isValid(user_i
d)) return res. status (400) .json({ message
: "Invalid user id"
```

```
const activity = await Activity.
  create({ user id, action
```

```
res. status( 201).j message: "Activity
  created successfully" , activity
```

```

    } catch (error) { res.
      status(500) .json({
        message: "Error creating activity" , error: error.
        message

```

```

const getActivities = async (req, res) => {
  try { const { user_id } req. params;

    if (
      !mongoose.Types.ObjectId.isValid(user_id)
    ) return res. status(400) .json(message .
      • "Invalid user id"

```

```

    const activities = await Activity
      . find({ user_id } )
      . sort({ timestamp: -1 } );
    res.json(activities);
  } catch (error) { res. status(500) .json(
    message : "Error getting activities" ,
    error: error. message

```

```

module. exports = { createActivity, getActivities } ;
routes/userRoutes . j s
const express = require( "express " ) ;

```

```

const router = express. Router();
const { createUser, getUser } =require( " . . /
controllers/userController" ) ;

```

```

router . post( " / users " , createUser) ;
router . get( " / users/ : id", getUser);

```

```

module. exports = router;
routes/ activityRoutes . j s
const express = require( " express");

```

```

const router = express.Router();
const { createActivity, getActivities } = require(
  "../controllers/activityController" );

router . post( "/" + activities , createActivity)
; router . get ( "/" + activities/ : user_id" ,
getActivities);
module. exports = router; app. js
const express = require("express");

const app = express();
require("../db"); app.

app.use(express.json());

const activityRoutes = require( "/" + routes/ activityRoutes " ); const
userRoutes = require( "/" + routes/userRoutes " );

app. use("/", activityRoutes) ; app.
use("/", userRoutes);

app. listen(5001, () => {console.log("User service running
on port 5001");

```